RRC-Raptor-V4H

- Switch

V0.0.5

Contents

LIST OF FIGURES

RETRONIX
The Solutions for Connected World

i. VERSION LIST

Description	Version	Date	Author
Switch	V0.0.1	2024/08/30	KevinLin
Modify 4.1.1(Page11)、4.1.4(Page14)、4.4.2(Page20)、4.4.3(Page21)、4.4.6(Page23)、4.6.2(Page27)、	V0.0.2	2024/10/30	KevinLin
File attribute changes	V0.0.3	2025/01/22	kevinlin
For Carrier Board v1.1 Modify 2.(Page8)、3.(Page9)、4.(Page10)、4.1.(Page14)、5.(Page31)、6.(Page32) Delete 4.4.4.SW21(page21) Add 4.4.4.SW46(page22)、4.6.5.SW47(page29)、4.6.6.SW48(page29)	V0.0.4	2025/02/17	kevinlin
Modify SW20, SW25	V0.0.5	2025/03/26	HowardChao

ii. HASHTAG

#V4H

#Switch

#DipSwitch

#Button

#Variable Voltage

#CB V1.1

1. Overview

Target：

DIP switches, buttons and variable voltage distribution, introduction and initial settings in Raptor-V4H.

2. Configuration Diagram

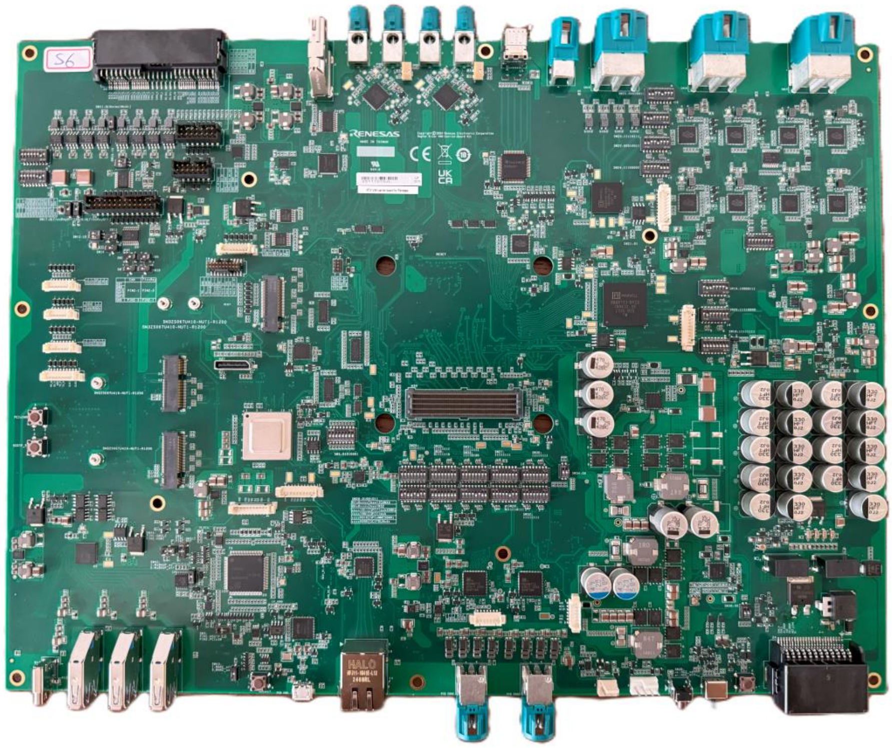

Figure 1. Top View

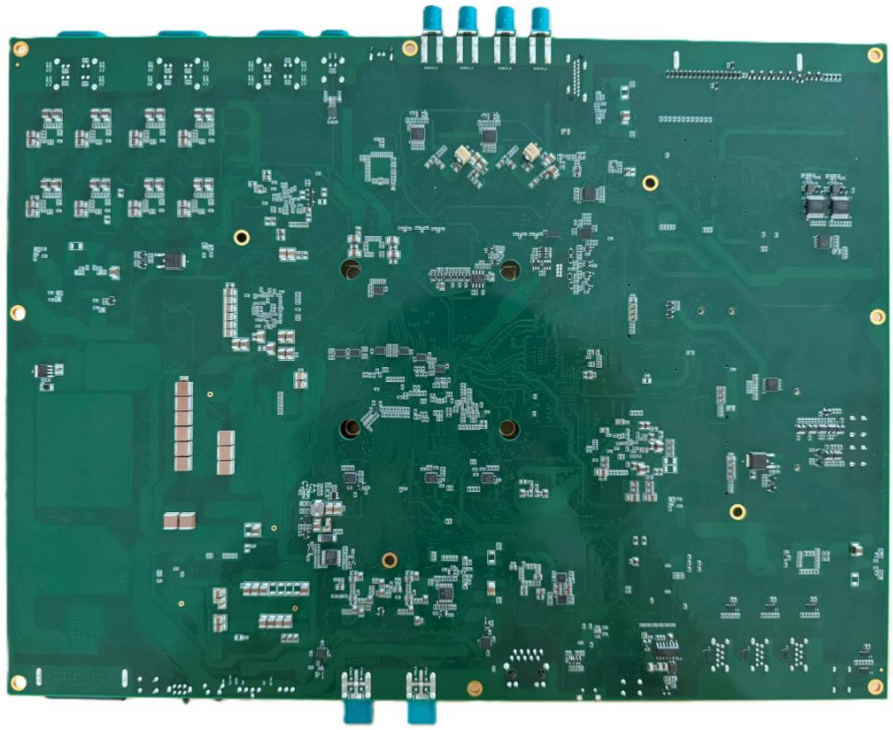

Figure 2. Bottom View

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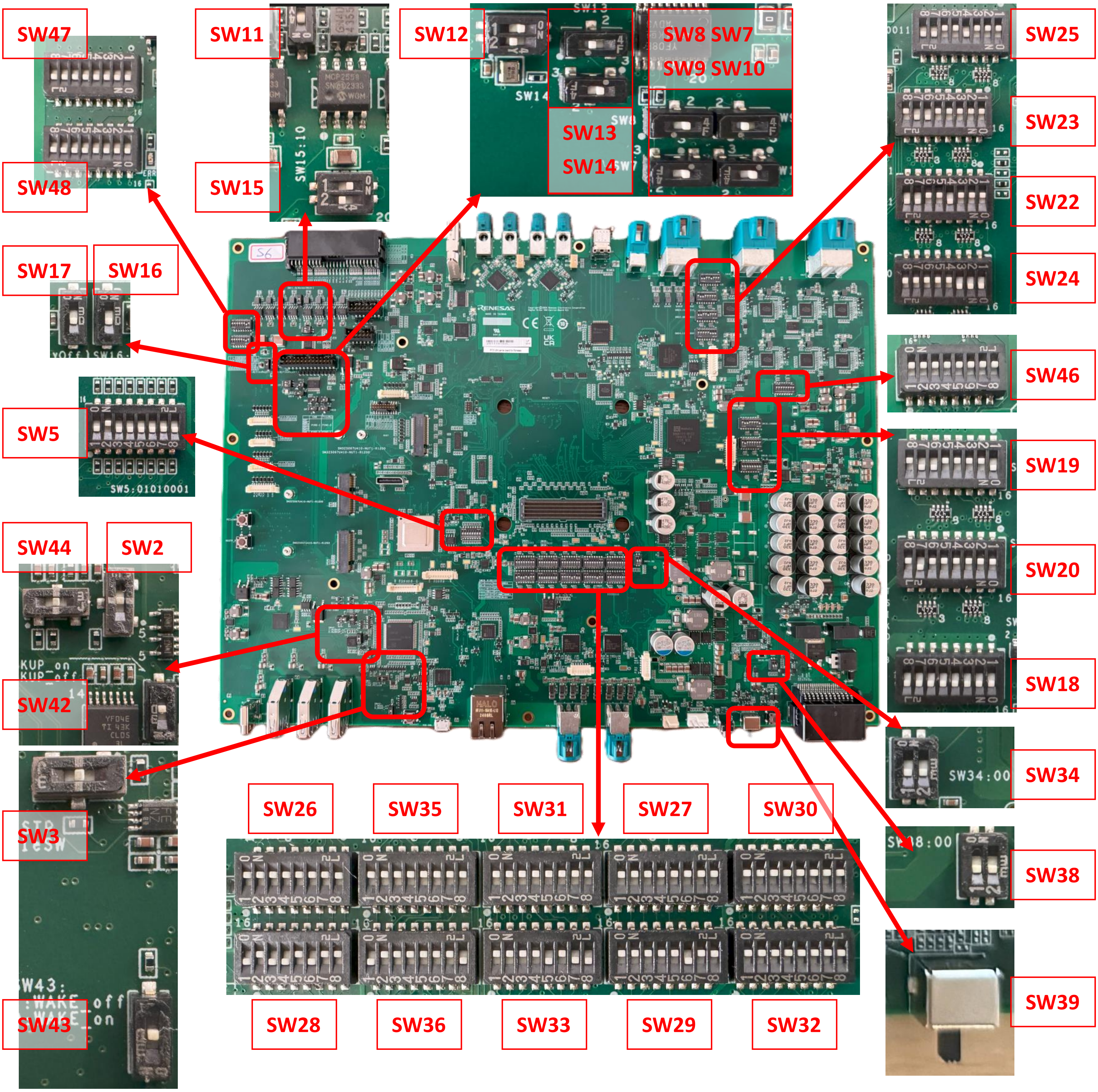
3. DIP SWITCH



4. SWITCH

Default Setting :

◆ Boot

SW26	SW35	SW31	SW27	SW30
1111 1111	**1111 1111**	**1111 1111**	**1111 1111**	**1111 1111**
SW28	SW36	SW33	SW29	SW32
0100 1011	**0111 1110**	**1111 0100**	**1111 1011**	**1111 1111**

◆ PCIe

SW5	SW3
01000001	**center**

◆ SPI

SW34	SW38
00	**00**

◆ Ethernet

SW18	SW19	SW20	SW22
1111 1111	**1000 0110**	**0000 0010**	**0001 0011**
SW23	SW24	SW25	SW46
0000 0110	**1110 0000**	**0001 0011**	**1111 1111**

◆ Display

SW2	SW44
center	**center**

◆ CANFD & FlexRay

SW7	SW8	SW9	SW10	SW13	SW14
PIN 2 – 1	**PIN 2 – 3**	**PIN 2 – 1**	**PIN 2 – 1**	**PIN 2 – 1**	**PIN 2 – 1**

SW11	SW12	SW15	SW16	SW17	SW47	SW48
1	**10**	**10**	**0**	**0**	**1111 1111**	**1111 1111**

◆ MCU & Power

SW39	SW42	SW43
0	**0**	**0**

4.1. BOOT MODE SWITCH

4.1.1. SW26 & SW28

SW26	1:MD7	2:MD6	3:X	4:MD8	5:MD4	6:MD3	7:MD2	8:MD1
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	PullHigh	PullHigh	X	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh
OFF	Open	Open	X	Open	Open	Open	Open	Open
SW28	1:MD7	2:MD6	3:X	4:MD8	5:MD4	6:MD3	7:MD2	8:MD1
DEFAULT	**OFF**	**ON**	**OFF**	**OFF**	**ON**	**OFF**	**ON**	**ON**
ON	GND	GND	X	GND	GND	GND	GND	GND
OFF	Open	Open	X	Open	Open	Open	Open	Open

◆ [SW28]

1:MD7	2:MD6	Selection of Master Boot Processor
OFF	**ON**	**Booted through ICUMAX**
OFF	OFF	Booted through Cortex-R52
Other than above		Setting prohibited

4:MD8	Selection mode
ON	Setting prohibited
OFF	**Use this setting**

5:MD4	6:MD3	7:MD2	8:MD1	Selection of Boot Device
ON	**OFF**	**ON**	**ON**	**Serial Flash boot at single read 40MHz using DMA**
ON	OFF	OFF	ON	Serial Flash boot at DDR read 80MHz using DMA
OFF	OFF	ON	OFF	eMMC boot at 50MHz x8 bus widths using DMA
OFF	OFF	OFF	OFF	SCIF downloading mode
Other than above				Setting prohibited

4.1.2. SW35 & SW36

SW35	1:MD5	2:MD20	3:MD11	4:MD10	5:MD35	6:MDT0	7:MD21	8:X
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X
OFF	Open	Open	Open	Open	Open	Open	Open	X
SW36	1:MD5	2:MD35	3:MDT0	4:MD21	5:MD20	6:MD3	7:MD2	8:X
DEFAULT	**OFF**	**ON**	**ON**	**ON**	**OFF**	**ON**	**OFF**	**OFF**
ON	GND	GND	GND	GND	GND	GND	GND	X
OFF	Open	Open	Open	Open	Open	Open	Open	X

◆ [SW36]

1:MD5	2:MD35	Selection mode
ON	-	Setting prohibited
OFF	-	**Normal boot**
-	**ON**	**Use this setting**
-	OFF	Setting prohibited

3:MDT0	4:MD21	5:MD20	6:MD11	7:MD10	Main_JTAG	
ON	ON	ON	ON	ON	-	Normal
ON	ON	ON	OFF	ON	ICUMX JTAG	Normal
ON	ON	ON	OFF	OFF	ICUMX LPD	Normal
ON	ON	OFF	ON	ON	ICUMX JTAG	Core Sight
ON	**ON**	**OFF**	**ON**	**OFF**	**ICUMX LPD**	**Core Sight**
ON	OFF	ON	ON	ON	Core Sight	Normal
OFF	ON	ON	ON	ON	VDSP0	Normal
OFF	ON	ON	ON	OFF	VDSP1	Normal
OFF	ON	ON	OFF	ON	VDSP2	Normal
OFF	ON	ON	OFF	OFF	VDSP3	Normal
OFF	ON	OFF	ON	ON	Core Sight	VDSP0
OFF	ON	OFF	ON	OFF	Core Sight	VDSP1
OFF	ON	OFF	OFF	ON	Core Sight	VDSP2
OFF	ON	OFF	OFF	OFF	Core Sight	VDSP3
Other than above					Setting prohibited	

4.1.3. SW31 & SW33

SW31	1:MD14	2:MD13	3:MD29	4:MD32	5:MD31	6:X	7:MD16	8:X
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X	PullHigh	X
OFF	Open	Open	Open	Open	Open	X	Open	X
SW33	1:MD14	2:MD13	3:MD29	4:X	5:MD16	6:MD32	7:MD31	8:X
DEFAULT	**ON**	**ON**	**ON**	**ON**	**OFF**	**ON**	**OFF**	**OFF**
ON	GND	GND	GND	X	GND	GND	GND	X
OFF	Open	Open	Open	X	Open	Open	Open	X

◆ [SW33]

1:MD14	2:MD13	Selection of PLL Initial Multiplication Ratio
ON	**ON**	**EXTAL input = 16.55 MHz, EXTAL Divider = x1/1**
ON	OFF	EXTEL input = 20.00 MHz, EXTAL Divider = x1/1
OFF	ON	Setting prohibited
OFF	OFF	EXTAL input = 33.33 MHz, EXTAL Divider = x1/2

3:MD29	5:MD16	Selection mode
ON	-	**Selects EXTALR as RCLK reference**
OFF	-	Selects on chip oscillator (OCO) as RCLK reference
-	ON	Setting prohibited
-	**OFF**	**Use this setting**

6:MD32	7:MD31	Selects SCIF or HSCIF downloading mode
ON	ON	SCIF download
ON	**OFF**	**HSCIF download 921600bps**
OFF	ON	HSCIF download 1843200bps
Other than above		Setting prohibited

4.1.4. SW27 & SW29

SW27	1:MD0	2:MD19	3:MD17	4:MD27	5:MD22	6:X	7:X	8:X
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X	X	X
OFF	Open	Open	Open	Open	Open	X	X	X
SW29	1:MD0	2:MD19	3:MD17	4:MD27	5:MD22	6:X	7:X	8:X
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	GND	GND	GND	GND	GND	X	X	X
OFF	Open	Open	Open	Open	Open	X	X	X

- [SW29]

4:MD0	Selection mode
ON	**Free-running mode**
OFF	Step-up mode

1:MD19	2:MD17	Setting of pin multiplexing of DDR phy
ON	**ON**	**6400 Mbps/pin for LPDDR5**
ON	OFF	6000 Mbps/pin for LPDDR5
OFF	ON	5500 Mbps/pin for LPDDR5
OFF	OFF	4800 Mbps/pin for LPDDR5

- [SW29]

1:MD27	2:MD22	Setting of pin multiplexing of DDR phy
ON	**ON**	**LPDDR5**
Other than above		Setting prohibited

4.1.5. SW30 & SW32

SW30	1:MD37	2:MD36	3:MD25	4:MD33	5:MD24	6:MD18	7:MD23	8:MD9
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh
OFF	Open	Open	Open	Open	Open	Open	Open	Open
SW32	1:MD37	2:MD36	3:MD25	4:MD23	5:MD33	6:MD24	7:MD18	8:MD9
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	GND	GND	GND	GND	GND	GND	GND	GND
OFF	Open	Open	Open	Open	Open	Open	Open	Open

- [SW32]

1:MD37	2:MD36	Selects PLL1 operation mode
ON	**ON**	**Integer mode (SSCG OFF)**
ON	OFF	Fractional mode (SSCG OFF)
OFF	ON	Reserved
OFF	OFF	Fractional mode, SSCG (Center spread ±1.0%)

4:MD23	5:MD33	6:MD24	7:MD18	Selection mode
ON	-	-	-	**Use this setting**
OFF	-	-	-	Setting prohibited
-	**ON**	-	-	**Use this setting**
-	OFF	-	-	Setting prohibited
-	-	**ON**	-	**Use this setting**
-	-	OFF	-	Setting prohibited
-	-	-	**ON**	**Use this setting**
-	-	-	OFF	Setting prohibited

3:MD25	8:MD9	Selection mode
ON	-	**Field BIST is not activated**
OFF	-	Field BIST is activated
-	**ON**	**The EXTAL pin is used for an oscillator**
-	OFF	The EXTAL and XTAL pins are used for a resonator

4.2. PCIE MODE SWITCH

4.2.1. SW5

SW5	1:PORT CFG0	2:PORT CFG1	3:PORT CFG2	4:JTAG_ SEL_L	5:I2C_A DDR[0]	6:I2C_A DDR[1]	7:I2C_A DDR[2]	8:SMBU S_EN_L
DEFAULT	**OFF**	**ON**	**OFF**	**OFF**	**OFF**	**OFF**	**OFF**	**ON**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:PORTCFG[0]	2:PORTCFG[1]	3:PORTCFG[2]	Function Mode
OFF	ON	ON	2Port - 16Lane
ON	OFF	ON	3Port - 16Lane
ON	OFF	ON	4Port - 16Lane
ON	ON	OFF	5Port - 16Lane
OFF	**ON**	**OFF**	**8Port - 16Lane**

5:I2C_ADDR[0]	6:I2C_ADDR[1]	7:I2C_ADDR[2]	I2C_ADDR = 0x1101[2:0]
ON	ON	ON	0x0: ADDR = 0x68
OFF	ON	ON	0x1: ADDR = 0x69
ON	OFF	ON	0x2: ADDR = 0x6A
OFF	OFF	ON	0x3: ADDR = 0x6B
ON	ON	OFF	0x4: ADDR = 0x6C
OFF	ON	OFF	0x5: ADDR = 0x6D
ON	OFF	OFF	0x6: ADDR = 0x6E
OFF	**OFF**	**OFF**	**0x7: ADDR = 0x6F**

4:JTAG_SEL_L	8:SMBUS_EN_L	Selection mode
ON	-	accessing PCIe PHY internal registers
OFF	**-**	**controlling switch's boundary scan**
-	**ON**	**SMBUS protocol is selected.**
-	OFF	I2C protocol is selected.

4.2.2. SW3

<table>
<tr><td>SW3</td><td>3:PCIE_MUX</td><td>2:PCIE_SEL</td><td>Connect</td></tr>
<tr><td>1-2</td><td>Don’t Care</td><td>H(P3.3V)</td><td>A to C
M.2 Key M(HSSSTP)</td></tr>
<tr><td>Center (DEFAULT)</td><td>Don’t Care</td><td>L</td><td>A to B
PCIe SW Connect</td></tr>
<tr><td rowspan="2">2-3</td><td>L</td><td>L</td><td>A to B
PCIe SW Connect</td></tr>
<tr><td>H</td><td>H</td><td>A to C
M.2 Key M(HSSSTP)</td></tr>
</table>

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4.3. SPI MODE SWITCH

4.3.1. SW34

SW34	1 : MSIOF2_SS2/MD25_SW	2 : GP1_25/SL_SW2_V
DEFAULT	**OFF**	**OFF**
ON	Pin LOW	Pin LOW
OFF	Pin HIGH(D3.3V_SPI)	Pin HIGH(D3.3V_SPI)

4.3.2. SW38

SW38	1 : AS_DOOR_SW	2 : GP1_25/SL_SW2_V
DEFAULT	**OFF**	**OFF**
ON	Pin LOW	Pin LOW
OFF	Pin HIGH(VIN_5V)	Pin HIGH(VIN_5V)

4.4. ETHERNET MODE SWITCH

4.4.1. SW18 (88Q6113)

SW18	1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	4:P9_SMODE[0]	5:P9_SMODE[1]	6:P10_SMODE[0]	7:P10_SMODE[1]	8:X
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	X
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X

1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	Selection RMU_SEL[2:0]
ON	**ON**	**ON**	**0x0: PORT 1 (SGMII)**
OFF	ON	ON	0x1: PORT 7 (xMII)
ON	OFF	ON	0x2: PORT 8 (xMII)
OFF	OFF	ON	0x3: PORT 9 (XFI)
OFF	OFF	OFF	0x7: RMU disable

4:P9_SMODE[0]	5:P9_SMODE[1]	Selection P9_SMODE[1:0]
ON	**ON**	**0x0: 1000BASE-X/SGMII**
OFF	ON	0x1: 5GBASE-R
ON	OFF	0x2: USXGMII
OFF	OFF	0x3: 10GBASE-R

6:P10_SMODE[0]	7:P10_SMODE[1]	Selection P10_SMODE[1:0]
ON	**ON**	**0x0: 1000BASE-X/SGMII**
OFF	ON	0x1: 5GBASE-R
ON	OFF	0x2: USXGMII
OFF	OFF	0x3: 10GBASE-R

4.4.2. SW19 (88Q6113)

SW19	1:P7 MODE0	2:P8 MODE0	3:P1 SMODE	4:P2 SMODE	5:P3 SMODE	6:C3 LED_RESV (P11_0)	7:P8 OUTEN_RESV (P11_1)	8:CPU_EN
DEFAULT	**ON**	**OFF**	**OFF**	**OFF**	**OFF**	**ON**	**ON**	**OFF**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:P7 MODE0	2:P8 MODE0	3:P1 SMODE	4:P2 SMODE	5:P3 SMODE	Selection mode
ON	-	-	-	-	**PORT 7 dsiable**
OFF	-	-	-	-	PORT 7 RGMII
-	ON	-	-	-	PORT 8 dsiable
-	**OFF**	-	-	-	**PORT 8 RGMII**
-	-	ON	-	-	100FX
-	-	**OFF**	-	-	**1000BASE-X/SGMII**
-	-	-	ON	-	100FX
-	-	-	**OFF**	-	**1000BASE-X/SGMII**
-	-	-	-	ON	100FX
-	-	-	-	**OFF**	**1000BASE-X/SGMII**

6:C3 LED_RESV (P11_0)	7:P8 OUTEN_RESV (P11_1)	Selection P11_SMODE[1:0]
ON	**ON**	**0x0: P11 PCIe**
OFF	ON	0x1: P11 PCIe
ON	OFF	0x2: P11 Eth port in 1000BASE-X/SGMII
OFF	OFF	0x3: P11 Eth port with 2500BASE-X

8:CPU_EN	Selection mode
ON	CPU Disable
OFF	**CPU Enable**

4.4.3. SW20 (88Q6113)

SW20	1:ADDR0	2:ADDR1	3:ADDR2	4:ADDR3	5:ACTIVE0	6:ACTIVE1	7:TEST	8:X
DEFAULT	**OFF**	**OFF**	**OFF**	**OFF**	**OFF**	**OFF**	**ON**	**OFF**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	X
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X

1:ADDR0	2:ADDR1	3:ADDR2	4:ADDR3	Selection ADDR[3:0]
ON	ON	ON	ON	MULTI-CHIP SMI ADDR 0xF
OFF	ON	ON	ON	MULTI-CHIP SMI ADDR 0xE
ON	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xD
OFF	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xC
ON	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xB
OFF	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xA
ON	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x9
OFF	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x8
ON	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x7
OFF	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x6
ON	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x5
OFF	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x4
ON	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x3
OFF	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x2
ON	OFF	OFF	OFF	MULTI-CHIP SMI ADDR 0x1
OFF	**OFF**	**OFF**	**OFF**	**SINGLE CHIP SMI ADDR 0x0**

5:ACTIVE0	6:ACTIVE1	Selection mode
ON	-	PORT 1 – 6 disable
OFF	-	**PORT 1 – 6 enable**
-	ON	PORT 7 – 10 disable
-	**OFF**	**PORT 7 – 10 enable**

7:TEST	Selection mode
ON	**DAP**
OFF	BSCAN

4.4.4. SW46 (88Q6113)

SW46	1:P1_RXD2	2:P2_RXD2	3:P3_RXD2	4:P4_RXD2	5:P5_RXD2	6:P6_RXD2	7:P9_RXD2	8:P10_RXD2
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:P1_RXD2	2:P2_RXD2	3:P3_RXD2	4:P4_RXD2	Selection mode
ON	-	-	-	**MASTER**
OFF	-	-	-	SLAVE
-	**ON**	-	-	**MASTER**
-	OFF	-	-	SLAVE
-	-	**ON**	-	**MASTER**
-	-	OFF	-	SLAVE
-	-	-	**ON**	**MASTER**
-	-	-	OFF	SLAVE

5:P5_RXD2	6:P6_RXD2	7:P9_RXD2	8:P10_RXD2	Selection mode
ON	-	-	-	**MASTER**
OFF	-	-	-	SLAVE
-	**ON**	-	-	**MASTER**
-	OFF	-	-	SLAVE
-	-	**ON**	-	**MASTER**
-	-	OFF	-	SLAVE
-	-	-	**ON**	**MASTER**
-	-	-	OFF	SLAVE

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4.4.5. SW22 (88Q5040)

SW22	1:P5_MODE[0]	2:P5_MODE[1]	3:P5_MODE[2]	4:P5_VDDOS	5:P6_MODE[0]	6:P6_MODE[1]	7:P6_MODE[2]	8:P6_VDDOS
DEFAULT	**OFF**	**OFF**	**OFF**	**ON**	**OFF**	**OFF**	**ON**	**ON**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:P5_MODE[0]	2:P5_MODE[1]	3:P5_MODE[2]	P5_MODE[2:0]
ON	ON	ON	0x0: MII MAC MODE
OFF	ON	ON	0x1: MII PHY
ON	OFF	ON	0x2: MII MAC
OFF	OFF	ON	0x3: RESV
ON	ON	OFF	0x4: RMII PHY
OFF	ON	OFF	0x5: RMII MAC
ON	OFF	OFF	0x6: DISABLE
OFF	**OFF**	**OFF**	**0x7: RGMII**

4:P5_VDDOS	Selection P5_VDDO
ON	**P5_VDDO = 2.5V**
OFF	P5_VDDO = 3.3V

5:P6_MODE[0]	6:P6_MODE[1]	7:P6_MODE[2]	P6_MODE[2:0]
ON	ON	ON	0x0: MII MAC MODE
OFF	ON	ON	0x1: MII PHY
ON	OFF	ON	0x2: MII MAC
OFF	**OFF**	**ON**	**0x3: RESV**
ON	ON	OFF	0x4: RMII PHY
OFF	ON	OFF	0x5: RMII MAC
ON	OFF	OFF	0x6: DISABLE
OFF	OFF	OFF	0x7: RGMII

8:P6_VDDOS	Selection P6_VDDO
ON	**P6_VDDO = 2.5V**
OFF	P6_VDDO = 3.3V

4.4.6. SW23 (88Q5040)

SW23	1:ADDR [0]n	2:ADDR [1]n	3:ADDR [2]n	4:ADDR [3]n	5:ACTIV E	6:GPIO_ VDDOS	7:SPI_V DDOS	8:CPU_ EN
DEFAULT	**OFF**	**OFF**	**OFF**	**OFF**	**OFF**	**ON**	**ON**	**OFF**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:ADDR[0]n	2:ADDR[1]n	3:ADDR[2]n	4:ADDR[3]n	Selection ADDR[3:0]
ON	ON	ON	ON	MULTI-CHIP SMI ADDR 0xF
OFF	ON	ON	ON	MULTI-CHIP SMI ADDR 0xE
ON	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xD
OFF	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xC
ON	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xB
OFF	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xA
ON	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x9
OFF	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x8
ON	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x7
OFF	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x6
ON	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x5
OFF	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x4
ON	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x3
OFF	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x2
ON	OFF	OFF	OFF	MULTI-CHIP SMI ADDR 0x1
OFF	**OFF**	**OFF**	**OFF**	**SINGLE CHIP SMI ADDR 0x0**

5:ACTIVE	6:GPIO_VDDOS	7:SPI_VDDOS	8:CPU_EN	Selection mode
ON	-	-	-	PORT 1 – 6 Disable
OFF	-	-	-	**PORT 1 – 6 Forward**
-	**ON**	-	-	**GPIO_VDDO = 2.5V**
-	OFF	-	-	GPIO_VDDO = 3.3V
-	-	**ON**	-	**SPI_VDDO = 2.5V**
-	-	OFF	-	SPI_VDDO = 3.3V
-	-	-	ON	CPU Disable
-	-	-	**OFF**	**CPU Enable**

4.4.7. SW24 (88Q5040)

SW24	1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	4:P1_MASTER	5:P2_MASTER	6:P3_MASTER	7:P4_MASTER	8:X
DEFAULT	**ON**	**ON**	**ON**	**OFF**	**OFF**	**OFF**	**OFF**	**OFF**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	X
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X

1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	RMU_SEL [2:0]
ON	**ON**	**ON**	**0x0: PORT 1 (100BT1)**
OFF	ON	ON	0x1: PORT 4 (100BT1/BTX/SGMII)
ON	OFF	ON	0x2: PORT 5 (xMII)
OFF	OFF	ON	0x3: RESV
ON	ON	OFF	0x4: RESV
OFF	ON	OFF	0x5: RESV
ON	OFF	OFF	0x6: RESV
OFF	OFF	OFF	0x7: RMU Disable

4:P1_MASTER	5:P2_MASTER	6:P3_MASTER	7:P4_MASTER	Selection mode
ON	-	-	-	SLAVE
OFF	-	-	-	**MASTER**
-	ON	-	-	SLAVE
-	**OFF**	-	-	**MASTER**
-	-	ON	-	SLAVE
-	-	**OFF**	-	**MASTER**
-	-	-	ON	SLAVE
-	-	-	**OFF**	**MASTER**

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4.4.8. SW25 (88Q5040)

SW25	1:MODE [0]	2:MODE [1]	3:MODE [2]	4:JTAG_ EN	5:SSPI_E N	6:SLEEP_ CAP_DIS	7:SSPI_M ODE[0]	8:SSPI_M ODE[1]
DEFAULT	**OFF**	**OFF**	**OFF**	**ON**	**OFF**	**OFF**	**ON**	**ON**
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:MODE[0]	2:MODE[1]	3:MODE[2]	PORT 4 MODE[2:0]	PORT 5 MODE[2:0]
ON	ON	ON	0x0: SGMII/1000X	0x0: RGMII
OFF	ON	ON	0x1: 100BASE-TX	0x1: 100BASE-FX
ON	OFF	ON	0x2: 100BASE-TX	0x2: SGMII/1000X
OFF	OFF	ON	0x3: 100BASE-TX	0x3: RGMII
ON	ON	OFF	0x4: 100BASE-FX	0x4: RGMII
OFF	ON	OFF	0x5: 100BASE-T1	0x5: 100BASE-FX
ON	OFF	OFF	0x6: 100BASE-T1	0x6: SGMII/1000X
OFF	**OFF**	**OFF**	**0x7: 100BASE-T1**	**0x7: RGMII**

4:JTAG_EN	5:SSPI_EN	6:SLEEP_CAP_DIS	Selection mode
ON	-	-	**NORMAL OPERATION**
OFF	-	-	JTAG ENABLE
-	ON	-	Slave SPI
-	**OFF**	-	**SMI_CPU**
-	-	ON	TC10 Enable
-	-	**OFF**	**TC10 Disable**

7:SSPI_MODE[0]	8:SSPI_MODE[1]	Selection SSPI_MODE[1:0]
ON	**ON**	**0x0: Clock Polarity = 0, Clock Phase = 0**
OFF	ON	0x1: Clock Polarity = 0, Clock Phase = 1
ON	OFF	0x2: Clock Polarity = 1, Clock Phase = 0
OFF	OFF	0x3: Clock Polarity = 1, Clock Phase = 1

4.5. DISPLAY MODE SWITCH

4.5.1. SW2 (DSI0)

SW2	3:SEL_DSI0	2:SELDSI0_R	Connect
1-2	Don't Care	H(D3.3V1_DSI)	A to C MAX96789 Connect
Center	**Don't Care**	**L**	**A to B** **SN65DSI86 Connect**
2-3	L	L	A to B SN65DSI86 Connect
	H	H	A to C MAX96789 Connect

4.5.2. SW44 (DSI1)

SW44	3:SEL_DSI1	2:SELDSI1_R	Connect
1-2	Don't Care	H(D3.3V1_DSI)	A to C MAX96789 Connect
Center	**Don't Care**	**L**	**A to B** **LT9611 Connect**
2-3	L	L	A to B LT9611 Connect
	H	H	A to C MAX96789 Connect

4.6. CAN MODE SWITCH

4.6.1. SW(7、8、9、10、13、14)

SW7	SW8	SW9	SW10	SW13	SW14	Selection mode
PIN 2 – 1	**PIN 2 – 3**	**PIN 2 – 1**	**PIN 2 – 1**	**PIN 2 – 1**	**PIN 2 – 1**	**CANFD1** **CANFD5**
PIN 2 – 3	PIN 2 – 1	PIN 2 – 3	PIN 2 – 3	PIN 2 – 3	PIN 2 – 3	FlexRay A FlexRay B

4.6.2. SW11

SW11 : CANFD5	1 : CAN_SILENT
DEFAULT	**ON**
ON	Short
OFF	Open

4.6.3. SW12

SW12	1 : RXD	2 : FXR_TXENB_D	Selection SW12
	ON	ON	Setting prohibited
DEFAULT	**ON**	**OFF**	**CANFD5 RXD**
	OFF	ON	FlexRay B TXEN
	OFF	OFF	Floating

4.6.4. SW15

SW15	1 : TXD	2 : FXR_TXENA_D	Selection SW15
	ON	ON	Setting prohibited
DEFAULT	**ON**	**OFF**	**CANFD5 TXD**
	OFF	ON	FlexRay A TXEN
	OFF	OFF	Floating

4.6.5. SW47

SW46	1:CANFD0_RX	2:CANFD0_TX	3:CANFD1_RX	4:CANFD1_TX	5:CANFD2_RX	6:CANFD2_TX	7:CANFD3_RX	8:CANFD3_TX
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	Short	Short	Short	Short	Short	Short	Short	Short
OFF	Open	Open	Open	Open	Open	Open	Open	Open

SW47	CANFD0~3 CAN_TX;CAN_RX
ON	MCP2558FD-H/SN
OFF	CAN BOARD CONNECTOR(CN26)

4.6.6. SW48

SW48	1:CANFD4_RX	2:CANFD4_TX	3:CANFD5_RX	4:CANFD5_TX	5:CANFD6_RX	6:CANFD6_TX	7:CANFD7_RX	8:CANFD7_TX
DEFAULT	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**
ON	Short	Short	Short	Short	Short	Short	Short	Short
OFF	Open	Open	Open	Open	Open	Open	Open	Open

SW48	CANFD4~7 CAN_TX;CAN_RX
ON	MCP2558FD-H/SN
OFF	CAN BOARD CONNECTOR(CN26)

4.7. FLEXRAY MODE SWITCH

4.7.1. SW16

SW16 : FlexRay Transceiver Channel A	1 : EN
DEFAULT	**OFF**
ON	Short (CAN_D3.3V_FLEX)
OFF	Open

4.7.2. SW17

SW17 : FlexRay Transceiver Channel B	1 : EN
DEFAULT	**OFF**
ON	Short (CAN_D3.3V_FLEX)
OFF	Open

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4.8. MCU MODE SWITCH

4.8.1. SW39

SW39	1 : ACC_SW
DEFAULT	**OFF**
ON	ACC_SW_MCU HIGH
OFF	ACC_SW_MCU LOW

4.8.2. SW42

SW42	1 : MCU_MXBKUPN
DEFAULT	**OFF**
ON	MxBKUP = 1
OFF	MxBKUP = 0

4.8.3. SW43

SW43	1 : TJA1463AT WAKE pin
DEFAULT	**OFF**
ON	WAKE OFF
OFF	WAKE ON

5. BUTTON

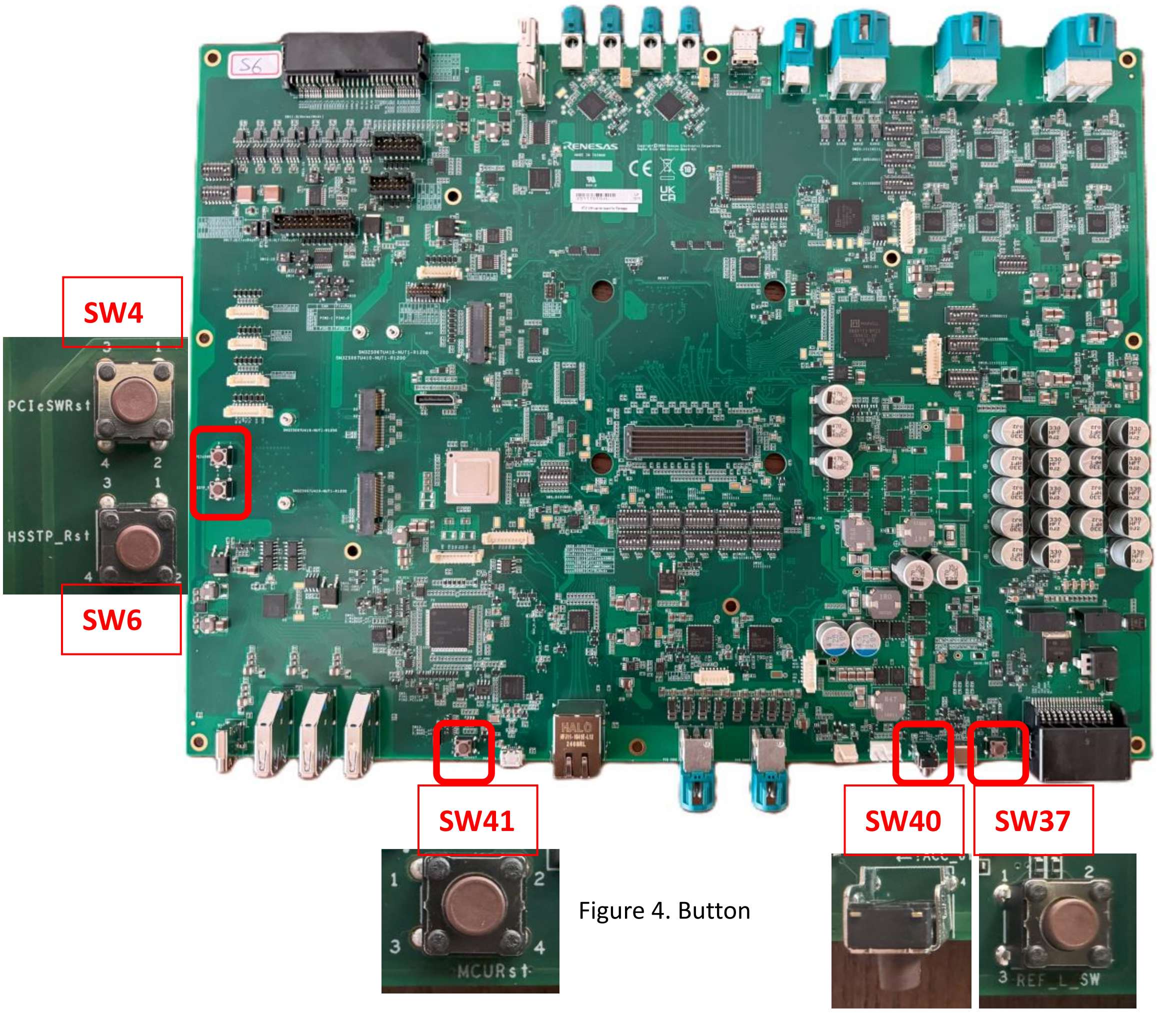


Figure 4. Button

SW40	Reset Switch
SW37	Reference line sw
SW41	MCU Reset
SW4	PCIe SW Reset
SW6	PCIe CN Reset

6. Variable Voltage

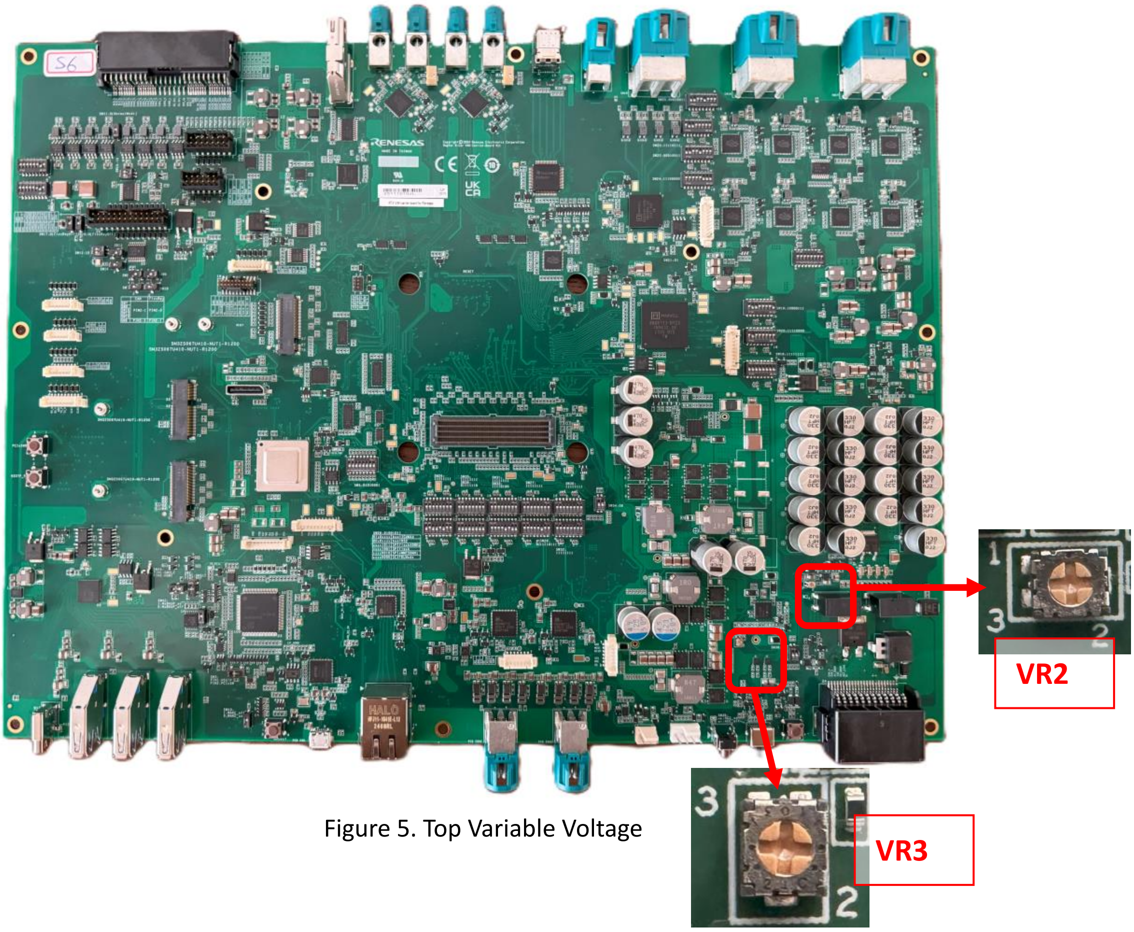
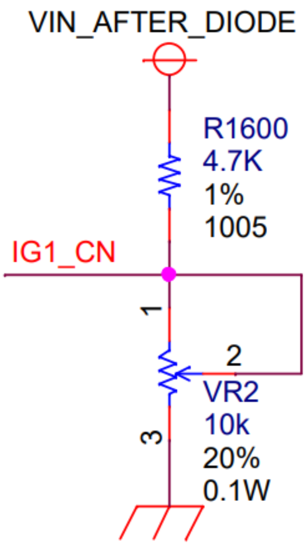
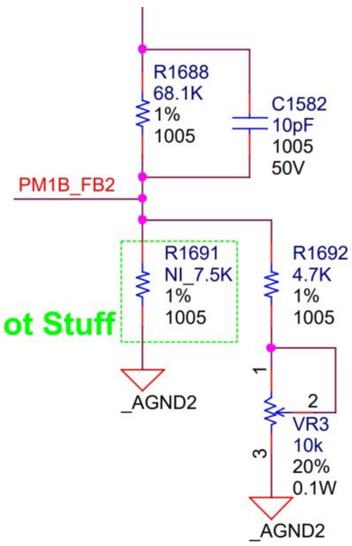


Figure 5. Top Variable Voltage

VR2	VR3
 	 R1692(4.7K), VR3 Mount/R1691(NI_7.5K) Dismount(default) Variable Voltage R1692(4.7K), VR3 Dismount/R1691(NI_7.5K) Mount Fixed Voltage 8.064V VOUT=(68.1k/7.5k+1)x0.8=8.064V

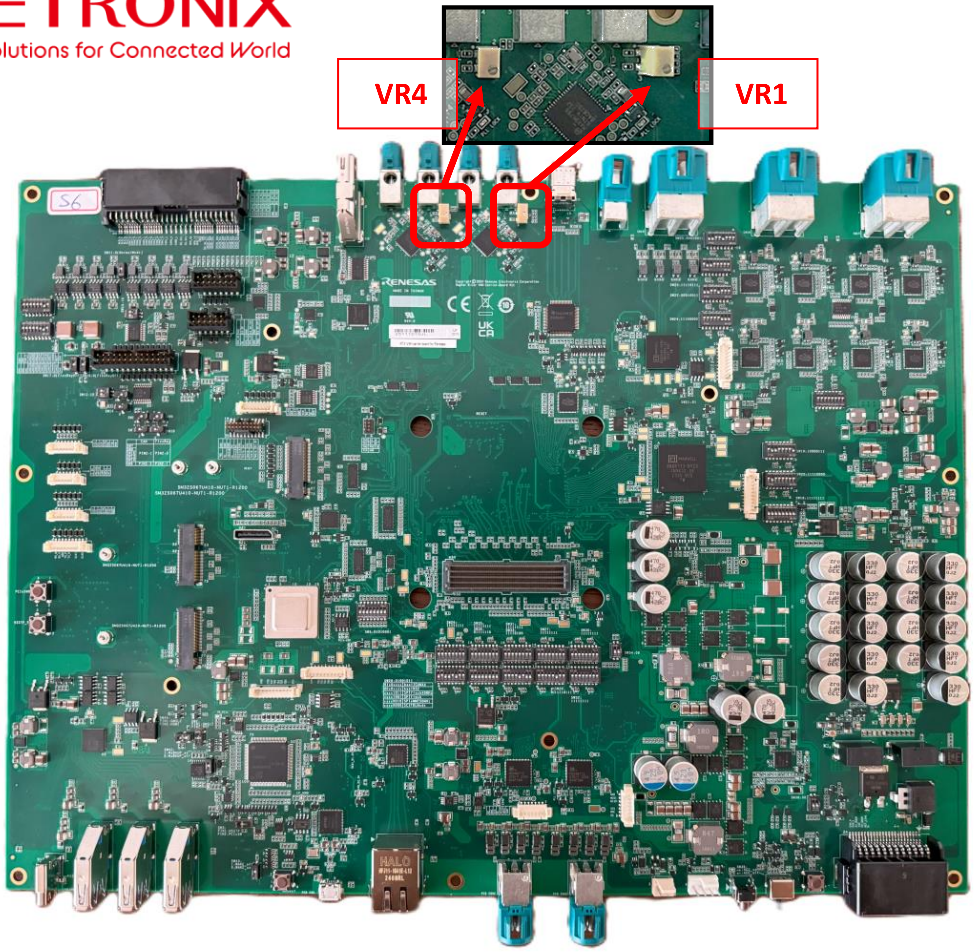
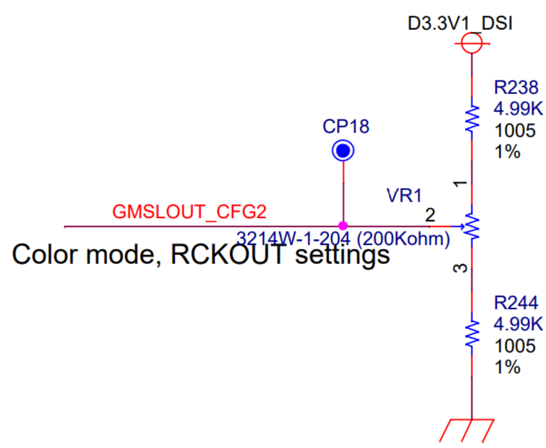
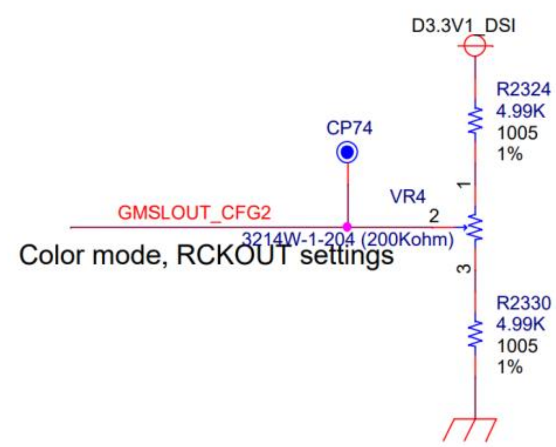


Figure 6. Bottom Variable Voltage

VR1 : U28 MAX96789GTN/V+	VR4 : U181 MAX96789GTN/V+
	

[Design Note] CFG2(CP9) voltage - setting table (@percentage of SPI_D3.3V)

MIN(%)	TYP(%)	MAX(%)	Color mode	RCLKOUT
0.0	0.0	11.7	24-bit mode	RCLKOUT disabled
16.9	20.2	23.6		RCLKOUT enabled
28.8	32.1	35.5	32-bit mode	RCLKOUT disabled
40.7	44.0	47.4		RCLKOUT enabled
52.6	56.0	59.3	27-bit mode (High BW mode)	RCLKOUT disabled
64.5	67.9	71.2		RCLKOUT enabled
76.4	79.8	83.1	Invalid	
88.3	100	100		

[Design Note] CFG2(CP9) voltage - setting table (@percentage of SPI_D3.3V)

MIN(%)	TYP(%)	MAX(%)	Color mode	RCLKOUT
0.0	0.0	11.7	24-bit mode	RCLKOUT disabled
16.9	20.2	23.6		RCLKOUT enabled
28.8	32.1	35.5	32-bit mode	RCLKOUT disabled
40.7	44.0	47.4		RCLKOUT enabled
52.6	56.0	59.3	27-bit mode (High BW mode)	RCLKOUT disabled
64.5	67.9	71.2		RCLKOUT enabled
76.4	79.8	83.1	Invalid	
88.3	100	100		

7. APPENDIX